21 **C** is the circle with equation $x^2 + y^2 = 80$

P is a point on **C**.
The coordinates of P are $(-4, k)$ where $k < 0$

Find an equation of the tangent to **C** at the point P.

$(-4)^2 + k^2 = 80$

1 mark
Substitute P in circle.

$k^2 = 64$; $k = -8$ because $k < 0$

1 mark
Select the negative square root.

Radius OP gradient $= (-8 - 0) \div (-4 - 0) = 2$

1 mark
The centre of $x^2 + y^2 = 80$ is (0, 0).

Tangent gradient $= -1/2$

1 mark
Tangent perpendicular to radius: negative reciprocal.

Through $(-4, -8)$: $y + 8 = -(1/2)(x + 4)$

$y = -(1/2)x - 10$

1 mark
Use the point-gradient equation, then simplify.

$y = -(1/2)x - 10$ .....

(Total for Question 21 is 5 marks)

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